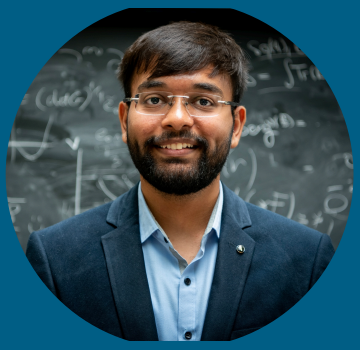
Vatsalya Vaibhav



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Links

- /Website
- /inspire-hep
- /linked-in
- /Twitter

About Me

I am a final year Ph.D. student at the Higgs Centre for Theoretical Physics at the University of Edinburgh. I am a cosmologist in spirit which means I am motivated to understand the Universe from the shortest length scales to the largest. My research includes a wide range of topics from quantum gravity and cosmology, lattice discretizations of fermions, to mathematical physics.

Education

Higgs Centre for Theoretical Physics, University of Edinburgh (Doctoral Degree)

Edinburgh September 2023 – August 2027

Degree: Doctor of Philosophy in Physics

Thesis: Fermions, in the sky, and on the lattice

Supervisor: Prof. Latham Boyle

Perimeter Institute for Theoretical Physics, University of Waterloo (Graduate Degree)

Waterloo Ontario Sep 2022 – June 2023

Degree: Master of Science

Perimeter Scholar International

Thesis: Quantized Supertwistors, Split-Octonions, and Unification

Supervisor: Prof. Latham Boyle

Indian Institute of Technology Kanpur (Undergraduate Degree)

Kanpur Aug 2018 – July 2022

Degree: Batchelor of Science (Physics)

Grade: 9 CGP

Publications

1. L. Boyle, **V. Vaibhav**, *"A New Approach to Chiral Gauge Theories on the Lattice"*, to appear soon.
2. V. Nenmeli, **V. Vaibhav**, *"Cosmological Perturbation Theory in the Bondi-Sachs Framework: Asymptotic Symmetries and Linear Cosmological Memory"*, to appear soon.
3. L. Boyle, N. Turok, **V. Vaibhav**, *"Might the radiation era extend back to the Big Bang? On dark matter production and the relic graviton background in quadratic gravity"*, arXiv:2609.11750.
4. L. Boyle, N. Turok, **V. Vaibhav**, *"Fixed points of classical gravity coupled with a Standard-Model-like theory"* arXiv:2509.09346.
5. P. Kaushik, **V. Vaibhav**, T.P. Singh, *"A proposed $E_8 \otimes E_8$ kinematic scaffolding for the standard model with pre-gravitation"*, Zeitschrift für Naturforschung A, 2026. https://doi.org/10.1515/zna-2026-0190, arXiv:2206.06911.
6. **V. Vaibhav** and T.P. Singh, *"Left-Right symmetric fermions and sterile neutrinos from complex split biquaternions and bioctonions"*, Adv.Appl.Clifford Algebras 33 (2023) 3, 32, arXiv: 2108.01858.
7. V. Bhatt, R. Mondal, **V Vaibhav**, and T.P. Singh, *"Majorana neutrinos, exceptional Jordan algebra, and mass ratios for charged fermions"* 2022 J. Phys. G: Nucl. Part. Phys. https://doi.org/10.1088/1361-6471/ac4c91, arXiv:2108.05787.

Conferences and Talks

1. **New dark matter production mechanism in quadratic gravity**, poster presentation at ERGE 2026, University of Sussex.
2. **Can the radiation era extend back to the Big Bang?**, seminar talk at Kobayashi Maskawa Institute for Theoretical Physics, Nagoya.
3. **A new approach to chiral gauge theories on the lattice**, talk at Frontiers of Lattice Fermions 2026, Yukawa Institute for Theoretical Physics, Kyoto.
4. **Can the radiation era extend back to the Big Bang?**, talk at Cosmological Frontiers in Fundamental Physics 2026, Perimeter Institute for Theoretical Physics, Waterloo Canada. https://pirsa.org/speaker/vatsalya-vaibhav-0
5. **A gravitational theory on the Lattice**, talk at *Lattice 2025, TIFR, Mumbai*.
6. **A physicist's proof of the Atiyah-Singer Index Theorem**, Student seminar, *Higgs Centre*, 2025.
7. **Fixed points of classical gravity coupled with a standard-model-like theory**, poster presentation at *Cosmological Frontiers in Cosmology, IAP Paris, 2025*.
8. **Fixed points of classical gravity coupled with a standard-model-like theory**, Gong show talk at *Tales of Gravity, Nottingham, 2025*.
9. **Asymptotically safe cosmology**, talk at *Young Theorists Forum (YTF), IAPP Durham, 2024*.
10. **Introduction to the exact renormalization group equation**, student seminar, *Higgs Centre, 2024*.
11. **Superfun with super-composition algebras and their application to supersymmetric spacetimes and the standard model**, student seminar, *Higgs Centre, 2023*.

Teaching

1. Symmetries of Particles and Fields (Fall 2023, 2024, 2025)
2. Quantum Field Theory (Fall 2023, 2024)
3. Gauge Theories in Particle Physics (Spring 2024, 2026)
4. Classical Electrodynamics (Spring 2025)
5. Symmetries in Quantum Mechanics (Spring 2025)

Scholarships and Awards

1. **Edinburgh Doctoral Scholarship**: prestigious scholarship from the University of Edinburgh
2. **Perimeter Scholars International Award**: prestigious scholarship from the University of Waterloo.
3. **National Standard Examination in Physics** state top 1% and qualified for the Indian National Physics Olympiad (**INPHO**)

Vatsalya Vaibhav

Mentoring

- Co-supervised a Master's project on boundary conditions for self-dual gravity at the Higgs Centre for Theoretical Physics.
- Mentored a summer reading project for first and second-year undergraduate students at IIT Kanpur on the book 'Spinors and Spacetime'.

Journal Refereeing

- The European Physical Journal C (EPJC): Particles and Fields.
- Nuclear Physics B: high energy physics, interface of particle physics and cosmology, quantum field theory, statistical systems, and mathematical physics.

Workshops and Internships

- Higgs Summer School on Theoretical Physics 2024-25-26
- British Universities Summer School on Theoretical and High Energy Physics 2024
- TRISEP, Perimeter Institute for Theoretical Physics 2023
- TIFR visiting student with professor T.P. Singh 2021
- Radio Astronomy Winter School (RAWSC) at Inter University Centre for Astronomy and Astrophysics (IUCAA) 2019

Volunteering Services

- Organizer for the theoretical and mathematical physics student seminars at the Higgs Centre. 2024-2026
- Member of the core team (academics) of the institute counselling service at IIT Kanpur.